

PRICING THE PROTOCOL AND DECOMPOSING THE RESIDUAL IN EXTREME-DARK IMAGE ENHANCEMENT

Heunseung Lim*, Jinhoi Kim, Youngmin Son, Jihyeon Ryu,
Seongwoong Kim, and Gichul Yoo

SK intellix

ABSTRACT

Scores for extreme-dark enhancement are not comparable across papers: four protocol choices—reference-mean rectification, the eight-bit rule, how PSNR is aggregated, and the rendition—move them, the largest of the three by ten decibels, by five among the paired-supervised. We rerun eighteen method–benchmark pairs under one protocol, price each choice where it applies, and decompose the residual with component-wise least-squares oracles, structured controls and band-wise reference correlation. Which component dominates varies with method and benchmark; the spatially varying low-frequency term is common to all, and the global term is a generalisation gap: its oracle gain is near one on a model’s training scenes, dispersed on new ones. The diagnosis prescribes one correction with a checkable precondition: rescale each output so its upper percentile reaches the ceiling where well-exposed references saturate. It gains where the precondition holds and harms where it fails; we report both.

Index Terms— Evaluation protocol, low-light image enhancement, error decomposition, band-wise correlation, image restoration

1. INTRODUCTION

Extreme-dark enhancement scores a short-exposure frame’s output against a long-exposure reference. Those scores are not comparable across papers: reference-mean rectification, the eight-bit rule, how PSNR is aggregated and the benchmark rendition each move a score, the largest difference by nearly ten decibels and by five among paired-supervised methods, so we rerun every baseline under one protocol and price each choice on the pairs it applies to. An improved score still does not identify the corrected error component.

We couple global, per-channel and spatial least-squares oracles with radial-band input–output correlation, applying the procedure to eighteen reproduced method–benchmark pairs on Sony sRGB and LOL. Noise and residual-donor controls probe spatial headroom; repeated captures support noise adjustment and frame-averaging tests. These measurements diagnose specified correction families.

Related work studies the effect of enhancement on downstream consumers [1, 2] and introduces Retinex decompositions [3, 4], zero-reference curves [5], normalising flows [6], colour and Fourier representations, signal-to-noise gating and transformers [7, 8, 9, 10, 11, 12], and diffusion priors [13, 14]. Brightness mismatch motivates mean-matching losses [15]; burst pipelines merge by frequency-dependent noise variance [16] or learn from dark bursts [17].

Oracle analyses have a history in denoising [18, 19], error decomposition in detection [20] and ring correlation in image assessment [21]. Closest to us is the least-squares photometric–structural split of [22], proved orthogonal there and made into a training loss by affine colour alignment, a photometric term containing our global and per-channel gains. We instead read that split off frozen checkpoints

as a diagnostic, and the correction it prescribes needs no reference at inference. We connect these band diagnostics to component-specific oracles, structured controls and repeated-capture adjustment, linking a score gap to a correction family whose predictability can be tested and, for the global term, exploited: one reference-free division whose precondition is measurable before use, worth +0.206 dB where it holds and costing 2.7 dB where it fails; we report both.

2. PROPOSED METHOD

The procedure combines component-specific least-squares (LS) oracles with input–output reference-correlation diagnostics. Let y be the model output and g the reference, in $[0, 255]$. PSNR is computed per frame and averaged; squared-error shares are pooled over frames. Shares use *unclipped* LS residuals. Reported global and channel-gain PSNR is clipped and rounded *after* the LS fit; the result is not an optimum over clipped estimators. Spatial oracle gains use unclipped PSNR so matched controls can be subtracted.

Gain terms. The global fit is $a^* = \langle y, g \rangle / \langle y, y \rangle$; fitting it separately per channel adds the colour term. The respective reductions in unclipped squared error define the global and additional channel shares. We fit a scalar per square block of side b , and separately an offset or affine map. Extra parameters improve even a noise fit, so each spatial gain over its whole-frame anchor is corrected by the gain of the same family against a structureless target of matched variance.

Band terms and aggregation. We Fourier-transform the luminance and group frequencies into radial bands. After fitting each input or output frame’s global brightness gain, let P_{ij} and G_{ij} be its and the reference’s band powers, and C_{ij} their real cross term. We report the pooled statistic

$$\rho_j = \frac{\sum_i C_{ij}}{\sqrt{(\sum_i P_{ij})(\sum_i G_{ij})}}. \quad (1)$$

Each frame contributes once, so scenes with more captures contribute more terms; this is not the mean of per-frame correlations. For a single frame, a real scalar band fit leaves $G_{ij}(1 - \rho_{ij}^2)$; the same identity holds for concatenated bands with one common scalar. Per-frame oracle PSNR is evaluated separately, not inferred from pooled ρ_j .

Correlation labels and noise correction. Compute ρ_j^{in} from the scalar-matched input by Eq. (1). Using fixed diagnostic thresholds, label a band *below floor* if $\rho_j^{\text{in}} < 0.20$; then label it *input advantage* if $\rho_j^{\text{in}} - \rho_j > 0.05$, and *no advantage* otherwise. Labelled energy shares sum the model’s pooled RGB residual energy over those bands. For repeated captures, estimate input noise power N_j from half the power of the difference between the first two captures of each scene–exposure group, scale it by each frame’s squared brightness gain and pool with the same frame weights as P_j , the mean input band power. The corrected input statistic is $\rho_j^{\text{in}} / \sqrt{1 - N_j / P_j}$. It assumes additive noise uncorrelated with the reference; $N_j / P_j > 0.9$ cells are unstable. Nonpositive signal power or a corrected magnitude

*Corresponding author: Heunseung Lim.

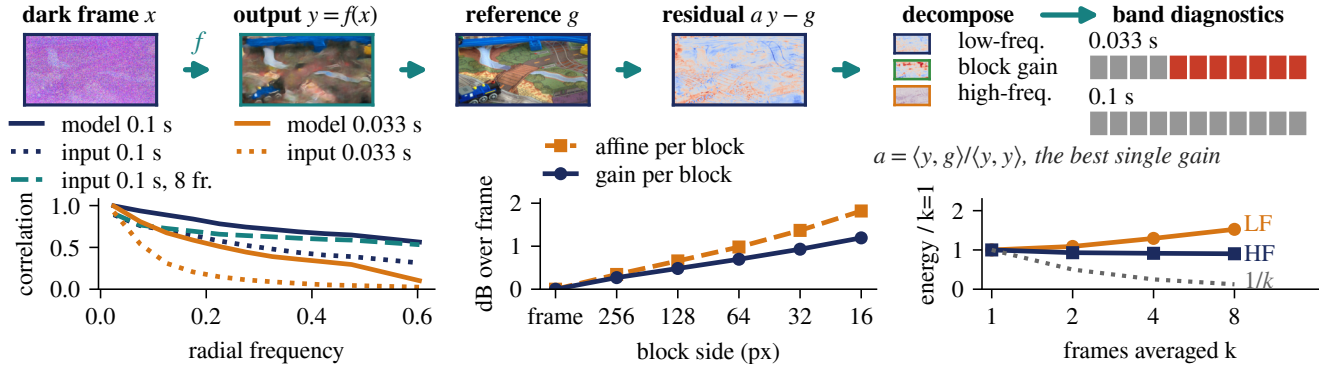


Fig. 1. Overview, worst 0.033 s Sony frame (input $\times 20$): the residual after the oracle global gain is split into low-frequency, per-block gain and high-frequency parts (real maps from this frame), and radial bands compare the input’s and output’s reference correlation (red: below floor; grey: no advantage; green: input advantage, absent on this frame). Labels describe the correlation test above. Bottom: benchmark measurements (LF/HF: low-/high-frequency error energy; the 0.1 s input curves use the 36 groups with at least eight frames).

above one invalidates the correction; such cells are not clipped into a label. The scene bootstrap draws 20000 resamples (seed 20260909) of scenes with equal probability and retains all frames of each, so scenes count equally and frames within a scene do not; the frame-weighted and scene-weighted gains are +0.206 and +0.211 dB. The correction is a sensitivity analysis under this noise model.

Predictability of the local term. Log input block mean, input variance and the reference-normalised output block mean predict within-frame, energy-weighted centred block gains, cross-fitted by scene so no frame is predicted from its own scene. Adding the log reference mean is a reference-assisted comparison unavailable at inference; a convolutional predictor tests richer features.

A highlight-anchored correction. The global term is an oracle, but the diagnosis makes it actionable under a checkable precondition. Well-exposed long-exposure references saturate at their upper percentiles: at $q = 99.9$, 100 percent of the Sony training references already reach the 8-bit ceiling, so the anchor is that ceiling, whose value needs no data. We therefore rescale each output by $1/Q_q(y)$, with Q_q its q th percentile, clamped to $[0.5, 2]$, a guard that never binds here; the rule needs no reference, calibration set or per-scene fitting. This is the one-sided, single-parameter case of percentile-based colour balance [23], the anchor that [24] applies to the input and that estimates atmospheric light [25]; we divide a trained output instead. The precondition is measurable on any benchmark by the fraction of training references saturated at q , and we report the correction where that fraction is high and where it is not. Measured on training references, it transfers: all 50 Sony test references saturate at 99.9 as well. We report its sensitivity to q rather than tuning it: $q = 95, 98, 99, 99.5, 99.9, 99.95$ and 99.99 give $-7.26, -3.30, -0.73, +0.35, +0.206, +0.16$ and $+0.08$ dB. Percentiles are non-decreasing in q and every training reference saturates at 99.9, so the precondition holds for every $q \geq 99.9$ —against 3 to 60 percent saturated at 95 to 99.5—and we take 99.9, the smallest on our grid at which every training reference saturates.

3. EXPERIMENTAL RESULTS

Setup. Every table and figure applies Sec. 2 to released checkpoints, training-free baselines or one in-house model. The Sony sRGB extreme-dark benchmark derives from SID [26] in the processed form prepared for SNR-Net [11] and released with Retinexformer [27]: 50 test scenes, 598 frames at 512×960 , each scored against its scene’s single long-exposure frame. LOL [28] (15 frames) with its v2 sets [29] is the standard low-light benchmark. The model analysed is the released Retinexformer checkpoint; inference is deterministic, so

spreads are over frames or scenes. We reproduce its reported score to within 0.002 dB with our own metric implementation (Sec. 4).

Controlled diagnostic validation. On 50 Sony references, the procedure separates a global gain from a smooth spatial gain of identical per-image MSE (global least squares removes 100 versus 76.6 percent of the error), and a reversible circular shift preserves power yet gives high-band correlation -0.0003 : low correlation is not information loss.

Four protocol choices move the score by decibels. First, reference-mean rectification sets the output’s brightness from the reference [30]: over the 18 method–benchmark pairs, it is worth -0.26 to $+9.81$ dB, the top end on a training-free baseline and $+1.16$ to $+4.90$ across the 8 pairs whose model was trained on paired references. One method recommends against the setting while another reports its headline with it, so the two are incomparable. Our anchor is the 8-bit ceiling, not a statistic of the evaluated scene. Second, the 8-bit rule: one pipeline truncates and another rounds; switching it alone is worth 0.031 to 0.110 dB on the three LOL checkpoints. Third, how PSNR is aggregated: the frame mean exceeds the PSNR of the pooled MSE, necessarily by AM–GM [31], and here by 0.82 to 3.31 dB. Fourth, the dataset variant: the Sony set is distributed in at least two renditions whose transfer functions differ [27, 7, 8], one built with no gamma curve, and the checkpoint of [7] scores 5.29 dB on the other, an absolute score rather than a difference, against the 22.904 reported on its own.

Existing methods under one protocol, and the proposed decomposition applied to each. Table 1 scores, with rounding and no rectification, 11 methods on LOL and 7 on Sony: CLAHE [32], ZeroDCE++ [5], SCI [33], an in-house self-supervised Retinex model (AR), LightenDiffusion [14] (unpaired; no Sony results reported), and reference-trained networks (URetinex-Net [4], SNR-Net [11], LLFormer [12], the diffusion model GSAD [13], Retinexformer, CIDNet [7]); the rightmost four columns are Sec. 2 applied to each method’s own residual; Fig. 2 shows one frame. On LOL, six methods clear 20 dB unrectified, the rest staying below 20 dB; on Sony only the two trained on it do, and LightenDiffusion leaves 24 percent of its error in the per-channel term, a colour cast. Across those methods (bounds rounded outward from the unrounded shares) the global term takes 41–89 percent of the error on LOL and 14–42 on Sony, so which term dominates varies with method and benchmark, while the block oracle stays at 0.8–3.0 dB on LOL and 0.7–1.8 dB on Sony: the spatially varying low-frequency term is common to all.

Global tone and colour: bounded LS corrections. On Sony, an oracle global gain and per-channel gain account for 27.0 and 8.8 per-

Table 1. Existing methods under one protocol (rounding, no rectification) and the decomposition of each residual (rightmost four columns): PSNR in dB (mean $\pm$ sd over frames; 15 on LOL, 598 on Sony), SSIM, squared-error shares in percent (pooled over frames) removed by the global gain and the per-channel gain, and the remainder, and Δ_{16} in dB (frame mean), the control-corrected 16-pixel block-gain oracle over the whole-frame fit. Diffusion models use one fixed-seed sample; URetinex-Net its fixed ratio; GSAD’s published 27.84 is rectified. Each block ends with the correction of Sec. 2 on that block’s Retinexformer row (Table 3 lists four Sony models); it is prescribed only where the precondition holds, so the LOL row is the control for failure. Best PSNR per benchmark in bold, second marked $\ddagger$; SSIM is not marked; its Sony margin is one printed digit. A per-block gain absorbs global scale, so Δ_{16} is unchanged.

LOL							Sony						
method	PSNR	SSIM	gl.	ch.	res.	Δ_{16}	method	PSNR	SSIM	gl.	ch.	res.	Δ_{16}
Input	7.77 $\pm$ 2.62	0.195	88.2	0.7	11.2	3.15	Input	11.50 $\pm$ 2.82	0.150	21.7	4.6	73.7	1.14
CLAHE	9.13 $\pm$ 2.96	0.362	88.2	0.4	11.4	2.99	CLAHE	13.32 $\pm$ 3.08	0.223	26.2	4.1	69.7	1.22
Zero-DCE++	15.34 $\pm$ 4.66	0.567	75.7	3.4	20.9	0.88	Zero-DCE++	12.74 $\pm$ 3.32	0.138	28.4	6.3	65.3	1.27
SCI	14.85 $\pm$ 4.43	0.524	70.1	2.1	27.8	1.74	SCI	11.80 $\pm$ 3.22	0.138	41.4	4.4	54.3	1.02
AR (in-house)	11.04 $\pm$ 3.72	0.449	86.5	0.8	12.7	0.96	AR (in-house)	12.93 $\pm$ 3.33	0.148	15.0	3.5	81.5	0.75
LightenDiffusion	20.12 $\pm$ 3.09	0.813	62.7	3.6	33.7	1.87	LightenDiffusion	17.09 $\pm$ 3.54	0.461	27.3	23.9	48.8	1.71
URetinex-Net	19.86 $\pm$ 3.97	0.825	73.5	3.2	23.2	1.73	SNR-Net	22.87 $\pm$ 3.52	0.625	33.6	9.9	56.5	1.35
SNR-Net	24.61 $\pm$ 4.26 $\ddagger$	0.842	53.7	4.3	42.0	1.15	Retinexformer	24.44 $\pm$ 3.51 $\ddagger$	0.680	27.0	8.8	64.2	1.19
LLFormer	23.65 $\pm$ 4.63	0.818	63.9	3.0	33.2	1.13	+anc. (ours)	24.64$\pm$3.43	0.681	22.8	9.3	67.9	1.19
GSAD	22.81 $\pm$ 5.42	0.852	80.9	1.3	17.8	1.37							
Retinexformer	25.15$\pm$2.87	0.845	41.4	4.1	54.5	1.17							
CIDNet	23.61 $\pm$ 5.44	0.870	75.7	1.1	23.3	2.29							
+anc. (ours)	22.47 $\pm$ 4.73	0.831	77.7	1.6	20.7	1.17							

input ($\times 1.4$)	Zero-DCE++	SNR-Net	LightenDiff.	Retinexformer	+anc. (ours)	reference
7.6 dB	8.5 dB	16.1 dB	11.6 dB	17.7 dB	18.1 dB	

Fig. 2. Existing methods and our calibration on the 0.033 s Sony frame on which Retinexformer scores lowest: the input after its oracle global gain (label: score before it), four method outputs, the anchored Retinexformer output ($\times 1.045$, using no reference at all), and the reference. Bottom: the low-frequency part (below 0.10) of each residual after its own oracle global gain, on a common 8-bit range; every method leaves a spatially varying brightness field.

cent of Retinexformer’s squared error, taking the score from 24.438 to 26.486 dB; the affine map of [22] takes 45.8 percent, +0.89 dB more; the remaining 64.2 percent is the unclipped residual after the per-channel LS optimum, growing from 60.8 percent at 0.1 s to 73.0 at 0.033 s.

Amplitude controls expose weak band agreement. Output power falls with radial frequency (0.994 in the lowest band, 0.279 at the corner), and the output is closer to a blurred reference than to the reference: at $\sigma = 0.6$ it scores 26.661 dB, 0.882 dB above its gain-corrected score (25.779), consistent with distortion-oriented smoothing [34]. Rescaling each band to the reference power with its own phase moves the score by 0.440 dB and the least-squares per-band gain by 0.947 dB, while ρ_j falls from near unity to 0.102 at the corner for the shortest exposure.

Raw and noise-adjusted diagnostics. Raw input correlation is below the model’s in every band and exposure (Table 2); below-floor bands hold 24.3 percent of residual energy at 0.033 s, and noise adjustment labels 0.0, 4.5 and 3.0 percent as input advantage. The last two are corner bands, gaps 0.058 and 0.062, and the 0.04 s cell is unstable at $N_j/P_j = 0.955$. Scene resampling (2000 draws, seed 20260907) retains them in only 69.2% and 65.6% of resamples, so these labels are uncertain. A repeated-capture check fails at the lowest band at 0.04 and 0.1 s and was not run at 0.033 s, so noise adjustment remains a model-dependent sensitivity.

Input and output averaging change different components. We test

Table 2. Pooled band-wise reference correlation for Retinexformer on Sony, Eq. (1): output and input, each scalar brightness-matched per frame (all frames). $\dagger$ marks unstable input cells ($N_j/P_j > 0.9$).

band	0.033 s ($n = 28$)		0.04 s ($n = 180$)		0.1 s ($n = 390$)	
	model	input	model	input	model	input
0.00–0.05	0.995	0.923	0.996	0.888	0.997	0.891
0.05–0.10	0.809	0.541	0.858	0.609	0.937	0.756
0.10–0.15	0.674	0.310	0.753	0.454	0.888	0.699
0.15–0.20	0.587	0.207	0.672	0.346	0.839	0.632
0.20–0.25	0.511	0.148 $\dagger$	0.595	0.261	0.782	0.560
0.25–0.30	0.444	0.112 $\dagger$	0.529	0.206	0.743	0.503
0.30–0.35	0.390	0.089 $\dagger$	0.483	0.169	0.716	0.456
0.35–0.40	0.357	0.069 $\dagger$	0.449	0.138 $\dagger$	0.686	0.411
0.40–0.45	0.327	0.053 $\dagger$	0.417	0.114 $\dagger$	0.664	0.379
0.45–0.50	0.295	0.045 $\dagger$	0.392	0.102 $\dagger$	0.648	0.360
0.50–0.71	0.102	0.025 $\dagger$	0.275	0.071 $\dagger$	0.564	0.284

averaging k input frames. On the 50 scene–exposure groups with at least eight frames (36 at 0.1 s, 14 at 0.04 s), so that every k is scored on identical groups, this raises the low-frequency error energy after the oracle global gain monotonically, to 1.089, 1.294 and 1.524 times the single-frame value at $k = 2, 4$ and 8, where pure variance would have given $1/k$, while the high-frequency energy falls only to 0.903. On the 36 scenes at 0.1 s, the eight-frame mean’s raw score falls 4.57 dB while the input’s band correlation rises in every band, at the corner from 0.315 to 0.532: high-frequency agreement improves as low-frequency error grows. Averaging outputs instead of inputs gains

0.672 dB at $k = 8$; the noise-realisation share is 10.6 percent, 0.49 dB, under independence. A noise-variance frequency-domain merge [16] at $k = 8$ recovers 0.91 dB of what the mean loses, and re-injecting Gaussian noise does not reverse the growth.

What carries over to the second benchmark. Three findings carry over to LOL: the output again closer to a blurred reference; the low-frequency part again dominant, 67 to 70 percent of the error energy; and the block oracle again above a decibel (Δ_{16} in Table 1). On LOL, a control fitted to another image’s residual gives 1.664 and 0.858 dB, 73 and 73 percent of the oracle gains. Control: no LOL band shows an input advantage against the raw input for either model.

The dominant term is a spatially varying low-frequency field. The residual after the global gain is concentrated at low frequency: the low-pass part carries 64.4 percent of the error energy, of which a single per-frame constant explains only 8.8 percent. Fig. 1 prices it: one scalar per 16-pixel block gains 1.194 dB over the whole-frame fit, an offset per block 1.09 dB and an affine map 1.819 dB, after subtracting the negligible structureless control. Residual-donor controls separate same-scene frames from other scenes at matched exposure and residual variance: same-scene donors give 1.209 dB and scene-excluded ones 1.035 dB ([0.984, 1.083]), so, much spatial fit remains for scene-excluded residuals, though the oracle measures a correction family’s fit, not an imaging cause. The reference-assisted three-statistic regression of Sec. 2, cross-fitted by scene, reaches $R^2 = -0.002$; adding the reference block mean, unavailable at inference, raises it to 0.693. A convolutional predictor on the input frame, split by scene, with failure fixed in advance at $R^2 < 0.1$ or a gain below 0.15 dB, reaches $R^2 = 0.019$ and 0.010 dB on the output, 0.8 percent of the oracle, with a label-shuffled control at or below zero and a single-batch overfit to $R^2 = 0.87$.

The global term is a generalisation gap, and what the prescribed correction recovers. The oracle global gain is worth 1.34 dB but averages to one per frame, so unconditionally a constant buys +0.01 dB. Its dispersion is a generalisation gap. Brightness lock-in on the training set and the rectification it forces are reported [35, 36]; the standard deviations here are 0.013 over training scenes, 0.104 over test scenes and 0.0024 within one scene and exposure. Pooled, Retinexformer gains +0.206 dB ([+0.049, +0.460], bootstrap $p = 0.0005$, scene t test 0.059; the signed rank 0.003 drops the 33 unchanged scenes, so it tests the 17 the rule acts on), SNR-Net +0.054 and LightenDiffusion +0.050 dB, Zero-DCE++ moves -0.018 at $p = 0.5263$, and SSIM moves from 0.680 to 0.681. Over the 10 pairs, Holm adjustment leaves 3 gains and 1 harm below 0.05 (Retinexformer 0.005). The precondition predicts the sign for seven of the ten pairs: 100 percent of the 181 Sony training references saturate at $q = 99.9$ against 25 percent of LOL’s 485, and there the rule harms 4 of 6 models, one by -2.88 dB, while a badly scaled one gains +3.01 dB. The rule is the identity wherever the output already reaches the ceiling: it multiplies 148 of 598 frames in 17 scenes, so the pooled figure averages over scenes it never changes. On the scenes it changes, the gain is +0.86 dB ([+0.32, +1.56], two-sided t 0.020 and sign test 0.013, unadjusted and post hoc, 14 of 17 positive; the best scene holds 35 percent of that mean, the top three 65 percent, and the best scene 48 percent of the scene mean), and the acting mean survives dropping its three best (+0.37 dB, t 0.021, leave-one-out in [+0.60, +0.94]). The scene mean does not: its top five scenes hold 88 percent of it, and dropping the best one and best five leaves +0.111 and +0.029 dB; nor does the pooled figure (86 percent, +0.122, +0.032). Over all frames of those scenes it is +0.56 dB per frame, +0.62 by scene. An oracle constant fitted to those frames buys +0.75 against +0.83 dB per frame; paired by scene, the difference is -0.05 [-0.89 , +0.84], 6 of 17 favouring the rule, which is therefore indistinguishable from a reference-fitted

Table 3. The prescribed correction on Sony. PSNR before and after, the gain with its scene bootstrap interval, the unadjusted p , and scenes improved.

model	PSNR	+anc.	Δ	95% CI	p	imp.
Retinexformer	24.44	24.64	+0.206	[+0.05,+0.46]	0.0005	14/50
SNR-Net	22.87	22.92	+0.054	[+0.01,+0.10]	0.0155	12/50
LightenDiffusion	17.09	17.14	+0.050	[+0.02,+0.10]	0.0005	18/50
Zero-DCE++	12.74	12.72	-0.018	[-0.08 ,+0.02]	0.5263	10/50

constant, to a resolution of 0.82 dB (95 percent of the effect), not better. Shuffling the multipliers among those frames leaves -0.01 dB (20 draws). Of the per-frame oracle gain (1.34, 2.12 dB) it recovers 15 and 39 percent. The worst case is -0.37 dB over the 50 scenes, -0.45 over the acting ones, -1.59 for one frame. Splitting all 598 frames by exposure—28, 180 and 390 from 14 nested in 29 nested in 50 scenes—gives +0.298, +0.249 and +0.179 dB, none negative, recombining to the pooled figure; scene t 0.055, 0.037 and 0.161 on the scene means of the same splits.

4. DISCUSSION

Reproduction. Baselines were rerun with independent PSNR/SSIM implementations agreeing with the reference ones to machine precision. CIDNet and Retinexformer match published scores within 0.021 dB except two cells, 0.081 and 0.060 dB, both rectified cells of one CIDNet variant; SNR-Net, LLFormer and URetinex-Net at its reference-derived ratio match LOL within 0.003 dB; the diffusion gaps are 0.13 and 0.34 dB, the latter inside a 0.57 dB seed spread that exceeds its anchored gain; Zero-DCE++ has no published LOL score.

What the decomposition implies for method design. The headroom on Sony is 1.34 dB for a global gain, 0.71 more per channel, the block and affine values of Fig. 1, 0.95 for per-band amplitude and about 0.49 for the noise share. Spend effort on the largest headroom after testing its predictors: the anchored correction does this for the global term, while the amplitude boost lowers PSNR and our low-frequency predictors leave most of the block oracle unused.

Limitations. All quantities describe the benchmarks’ 8-bit sRGB renditions. Quantisation and non-additive sensor noise [37] depart from the additive-noise model behind the noise correction: its check fails in the lowest band at 0.04 and 0.1 s, and $N_j/P_j > 0.9$ cells are reported as unstable. Correlation is distortion-oriented [34]: neither low correlation nor a failed predictor establishes the absence of information. The reference’s noise floor is treated as zero. Noise correction needs repeated captures, so it exists only on Sony. The band-wise test covers Retinexformer on Sony and two models on LOL; CIDNet’s Sony residual was not decomposed because its training variant uses another pixel convention. The correction is one division, neither uniform nor portable: it acts only on unsaturated outputs, harms most models where the precondition fails, and cannot be validated on training scenes, where the oracle gain is near one. Pooled band correlations can hide local disagreement, and the multi-frame result uses static scenes only.

5. CONCLUSION

Extreme-dark scores are protocol-bound: four choices move them, the largest difference by nearly ten decibels, so we reran eighteen pairs under one protocol and priced three of the choices as differences. Component-wise oracles, equal-error controls and band-wise diagnostics then say which term to attack: the low-frequency term is common to all and resists our predictors; the global term is a generalisation gap. From that follows one correction with a checkable precondition, worth +0.206 dB on Sony and costing 2.7 dB where it fails; both are reported. Code: github.com/HeunSeungLim/lowlight-decomp.

6. REFERENCES

- [1] Xu Wu, Zhihui Lai, Jie Zhou, Can Gao, Xianxu Hou, Ya-nan Zhang, and Linlin Shen, “Low-light enhancement effect on classification and detection: An empirical study,” *arXiv preprint arXiv:2409.14461*, 2024.
- [2] Mingjia Li, Hao Zhao, and Xiaojie Guo, “LIME-Eval: Rethinking low-light image enhancement evaluation via object detection,” *arXiv preprint arXiv:2410.08810*, 2024.
- [3] Yonghua Zhang, Jiawan Zhang, and Xiaojie Guo, “Kindling the darkness: A practical low-light image enhancer,” in *ACM Multimedia*, 2019, pp. 1632–1640.
- [4] Wenhui Wu, Jian Weng, Pingping Zhang, Xu Wang, Wenhan Yang, and Jianmin Jiang, “URetinex-Net: Retinex-based deep unfolding network for low-light image enhancement,” in *CVPR*, 2022, pp. 5901–5910.
- [5] Chongyi Li, Chunle Guo, and Chen Change Loy, “Learning to enhance low-light image via zero-reference deep curve estimation,” *IEEE TPAMI*, vol. 44, no. 8, pp. 4225–4238, 2022.
- [6] Yufei Wang, Renjie Wan, Wenhan Yang, Haoliang Li, Lap-Pui Chau, and Alex C. Kot, “Low-light image enhancement with normalizing flow,” in *AAAI*, 2022, vol. 36, pp. 2604–2612.
- [7] Qingsen Yan, Yixu Feng, Cheng Zhang, Guansong Pang, Kangbiao Shi, Peng Wu, Wei Dong, Jinqu Sun, and Yanning Zhang, “HVI: A new color space for low-light image enhancement,” in *CVPR*, 2025, pp. 5678–5687.
- [8] Qingsen Yan, Kangbiao Shi, Yixu Feng, Tao Hu, Peng Wu, Guansong Pang, and Yanning Zhang, “HVI-CIDNet+: Beyond extreme darkness for low-light image enhancement,” *arXiv preprint arXiv:2507.06814*, 2025.
- [9] Jie Huang, Yajing Liu, Feng Zhao, Keyu Yan, Jinghao Zhang, Yukun Huang, Man Zhou, and Zhiwei Xiong, “Deep Fourier-based exposure correction network with spatial-frequency interaction,” in *ECCV*, 2022, pp. 163–180.
- [10] Chenxi Wang, Hongjun Wu, and Zhi Jin, “FourLLIE: Boosting low-light image enhancement by Fourier frequency information,” in *ACM Multimedia*, 2023, pp. 7459–7469.
- [11] Xiaogang Xu, Ruixing Wang, Chi-Wing Fu, and Jiaya Jia, “SNR-aware low-light image enhancement,” in *CVPR*, 2022, pp. 17714–17724.
- [12] Tao Wang, Kaihao Zhang, Tianrun Shen, Wenhan Luo, Bjorn Stenger, and Tong Lu, “Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method,” in *AAAI*, 2023, vol. 37, pp. 2654–2662.
- [13] Jinhui Hou, Zhiyu Zhu, Junhui Hou, Hui Liu, Huanqiang Zeng, and Hui Yuan, “Global structure-aware diffusion process for low-light image enhancement,” in *NeurIPS*, 2023.
- [14] Hai Jiang, Ao Luo, Xiaohong Liu, Songchen Han, and Shuaicheng Liu, “LightenDiffusion: Unsupervised low-light image enhancement with latent-Retinex diffusion models,” in *ECCV*, 2024.
- [15] Jingxi Liao, Shijie Hao, Richang Hong, and Meng Wang, “GT-Mean loss: A simple yet effective solution for brightness mismatch in low-light image enhancement,” in *ICCV*, 2025.
- [16] Samuel W. Hasinoff, Dillon Sharlet, Ryan Geiss, Andrew Adams, Jonathan T. Barron, Florian Kainz, Jiawen Chen, and Marc Levoy, “Burst photography for high dynamic range and low-light imaging on mobile cameras,” *ACM TOG*, vol. 35, no. 6, pp. 192:1–192:12, 2016.
- [17] Ahmet Serdar Karadeniz, Erkut Erdem, and Aykut Erdem, “Burst photography for learning to enhance extremely dark images,” *IEEE TIP*, vol. 30, pp. 9372–9385, 2021.
- [18] Priyam Chatterjee and Peyman Milanfar, “Is denoising dead?,” *IEEE TIP*, vol. 19, no. 4, pp. 895–911, 2010.
- [19] Anat Levin and Boaz Nadler, “Natural image denoising: Optimality and inherent bounds,” in *CVPR*, 2011, pp. 2833–2840.
- [20] Derek Hoiem, Yodsawalai Chodpathumwan, and Qieyun Dai, “Diagnosing error in object detectors,” in *ECCV*, 2012, pp. 340–353.
- [21] J. Kaczmar-Michalska, N. R. Hajizadeh, A. J. Rzepiela, and S. F. Nørrelykke, “Image quality measurements and denoising using Fourier Ring Correlations,” *arXiv preprint arXiv:2201.03992*, 2022.
- [22] Mingjia Li, Tianle Du, Hainuo Wang, Qiming Hu, and Xiaojie Guo, “On the global photometric alignment for low-level vision,” *arXiv preprint arXiv:2604.08172*, 2026.
- [23] Nicolas Limare, Jose-Luis Lisani, Jean-Michel Morel, Ana Belén Petro, and Catalina Sbert, “Simplest color balance,” *Image Processing On Line*, vol. 1, pp. 297–315, 2011.
- [24] Peiyuan He, Hainuo Wang, Hengxing Liu, Mingjia Li, and Xiaojie Guo, “Internally referenced low-light enhancement,” *arXiv preprint arXiv:2605.28605*, 2026.
- [25] Kaiming He, Jian Sun, and Xiaoou Tang, “Single image haze removal using dark channel prior,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, 2011.
- [26] Chen Chen, Qifeng Chen, Jia Xu, and Vladlen Koltun, “Learning to see in the dark,” in *CVPR*, 2018, pp. 3291–3300.
- [27] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang, “Retinexformer: One-stage Retinex-based transformer for low-light image enhancement,” in *ICCV*, 2023.
- [28] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu, “Deep Retinex decomposition for low-light enhancement,” in *BMVC*, 2018.
- [29] Wenhan Yang, Wenjing Wang, Haofeng Huang, Shiqi Wang, and Jiaying Liu, “Sparse gradient regularized deep Retinex network for robust low-light image enhancement,” *IEEE TIP*, vol. 30, pp. 2072–2086, 2021.
- [30] Kanghao Chen, Guoqiang Liang, Hangyu Li, Yunfan Lu, and Lin Wang, “Evlight++: Low-light video enhancement with an event camera,” *arXiv preprint arXiv:2408.16254*, 2024.
- [31] Onur Keleş, M. Akın Yılmaz, A. Murat Tekalp, Cansu Korkmaz, and Zafer Dogan, “On the computation of PSNR for a set of images or video,” in *Picture Coding Symposium (PCS)*, 2021.
- [32] Karel Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, pp. 474–485. Academic Press, 1994.
- [33] Long Ma, Tengyu Ma, Risheng Liu, Xin Fan, and Zhongxuan Luo, “Toward fast, flexible, and robust low-light image enhancement,” in *CVPR*, 2022.
- [34] Yochai Blau and Tomer Michaeli, “The perception-distortion tradeoff,” in *CVPR*, 2018, pp. 6228–6237.
- [35] Hongru Han, Tingrui Guo, Liming Zhang, Yan Su, Qiwen Xu, and Zhuohua Ye, “Towards controllable low-light image enhancement: A continuous multi-illumination dataset and efficient state space framework,” *arXiv preprint arXiv:2603.25296*, 2026.
- [36] Maria Pilligua, David Serrano-Lozano, Pai Peng, Ramon Baldrich, Michael S. Brown, and Javier Vazquez-Corral, “Evaluating low-light image enhancement across multiple intensity levels,” *arXiv preprint arXiv:2511.15496*, 2025.
- [37] Kaixuan Wei, Ying Fu, Jiaolong Yang, and Hua Huang, “A physics-based noise formation model for extreme low-light raw denoising,” in *CVPR*, 2020, pp. 2758–2767.